SCHOOL OF ARCHITECTURE, COMPUTING & ENGINEERING

INDIVIDUAL CW B REPORT CN7029

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TITLE

“COMPARATIVE RESEARCH ON GPU ITS OPERATING SYSTEMS AND THEIR APPLICATION IN AWS OR AZURE CLOUD PLATFORMS”

INTRODUCTION

This report demonstrates the study of GPU usage supported by Linux GPU Operating system command-line tools and scrutinize their application within cloud computing platforms. It has included a comparison of AWS cloud services and Microsoft Azure. For the better explanation of services and core components of the comparison is best to be discussed. This report will be a brief description of Linux GPU based operations.

Linux is a major operating system for GPU-enabled conditions due to its resilience, stability, and wide-range support for command-line tools that are effectively used for GPU monitoring and performance management. These command-line tools enable system users to assess GPU utilization, detect errors, and optimize system performance. These tools genuinely ensure reliability and efficiency in the required workloads.

A practical cloud implementation depicts GPU-related operations. It concludes with an evaluation of the role of Linux GPU tools in GPU and cloud computing systems. Cloud computing platforms, for example Amazon web services (AWS) and Microsoft Azure, have integrated GPU. These platforms provide scalable, on-demand access to GPU resources, which allows companies to deploy highly GPU intensive GPU services.

This Coursework explains the study of GPU uses stems from Linux Gpu Operating system tools and examines their integration within AWS. The Aim of this report is to get an understanding of modern times GPU utilization with Linux in Cloud computing system and how linux made it more useful for users.

OVERVIEW OF GPU AND LINUX GPU OPERATING SYSTEM COMMAND-LINE TOOLS

Graphics processing units (GPUS) are specialized hardware components built to work parallelly processing tasks perfectly. Reasonably, the capacity to carry out multiple tasks, i.e. computations simultaneously, high-performance computing, and it provides higher performance for intensive tasks. Modern devices come with Gpu integrated with CPU for the best processes of graphical user interface in the latest application like games. It provides better performance without any doubt. Gpu contains many cores capable of doing multiple operations at a single time.

Linux is the commonly used OS due to its driver support. However, Linux stems from GPUs through open-source and private drivers. The kernel relates to GPUs through Direct Rendering Manager (DRM). Commonly GPU vendors on Linux are NVIDIA, AMD, and Intel. Linux allows users to configure GPU with command line tools which are necessary for cloud systems.

In cloud computing platforms such as AWS and Microsoft Azure, Linux GPU command-line tools are used for managing GPU-enabled virtual machines. The cloud-based engineers used it to check utilization of hardware in the system for managing the application and expenses as well.

Common Linux GPU commands are demonstrated in further detail which are beneficial to users for knowing their hardware presence in the system. Otherwise, users are not aware of the system requirements and details of the drivers. Altogether, these tools make it possible to manage GPU systems withing Linux environments.

These help to verify GPU availability, manage workloads, and ensure use of GPU resources. Major points covered: -

- 5 Linux GPU command line tools are used in GPU
- Comparison between Azure and AWS cloud platform
- Discussion of Linux GPU command-line tools used in GPU and Cloud computing

I have used Microsoft Azure for the deployment of the resource group and Identity access management. The fundamental components of GPU running can be managed by cloud-based computing infrastructures.

DEMONSTRATION HOW 5 LINUX GPU COMMAND LINE TOOLS ARE USED IN GPU

The following command in Linux command line operating system with their explanation shows information about GPU is connected to the system, which is helpful for users to check, maintain and manage their usage. The resilience and accessibility of Linux operating systems for the GPU command is best among the other operating systems, that's why Linux is the most used operating system for cloud computing.

1. **sudo lshw -C display**
   This command is to show Apple GPU recognized by Linux.
   My personal laptop does not have a NVidia GPU card, so I must use this specific GPU command to check the GPU properties of the system.

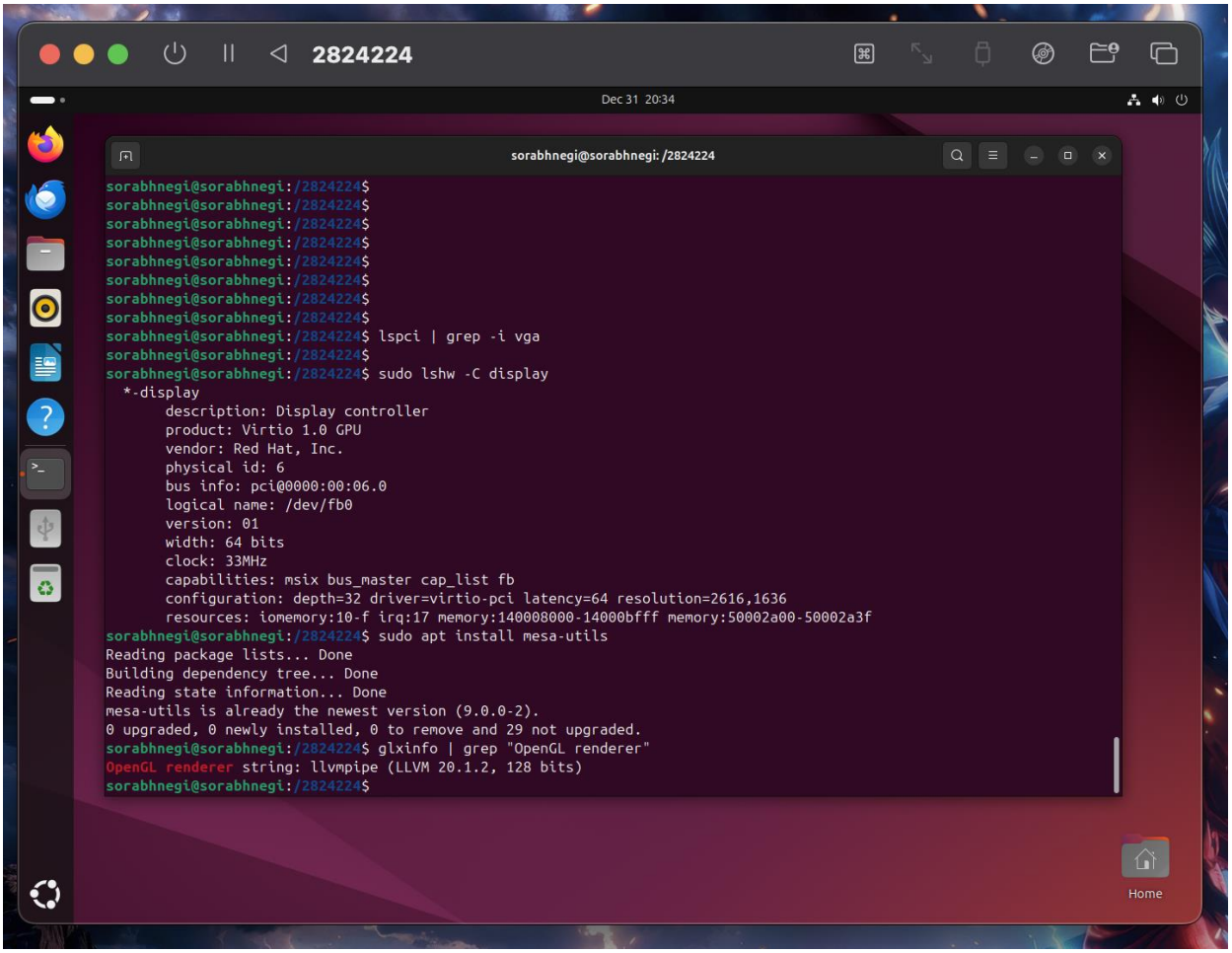


Figure 1

2. Sudo apt install mesa-utils
   This command is used to install the mesa utilities for the Gpu otherwise the following command will not run.

Glxinfo | grep “OpenGL renderer” and this command helps to check the information related to GLX information in the renderer.

This command confirms that the GPU driver is used by Apple.

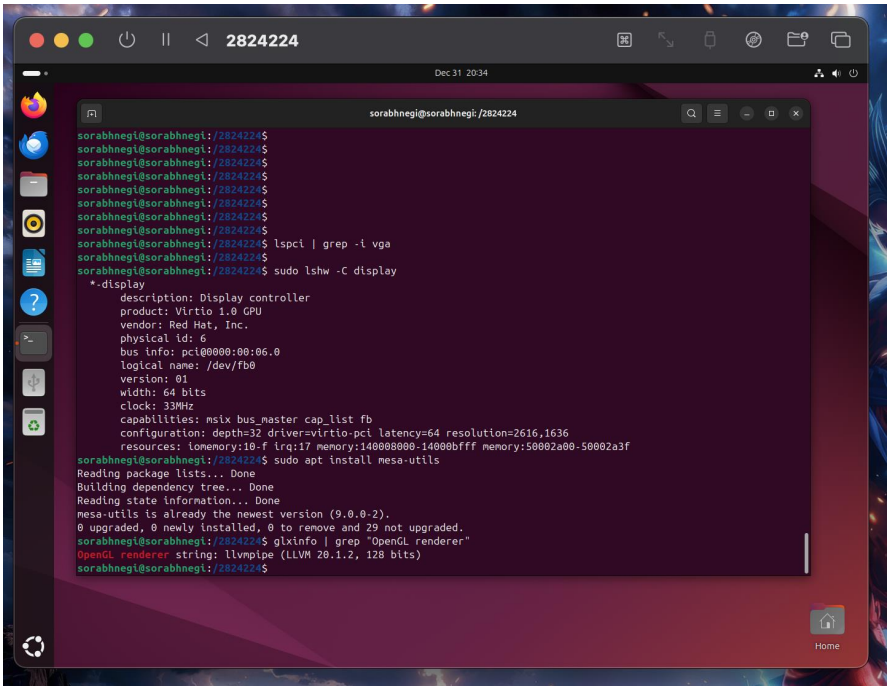


Figure 2

3. Sudo apt install vulkan-tools.

   The command is specifically used for the installation of the vulkan-tools. Which enables you to run the following command.

   Vulkaninfo | less this command makes it possible to check the Vulkan info install in the system.

This shows that Vulkan is available.

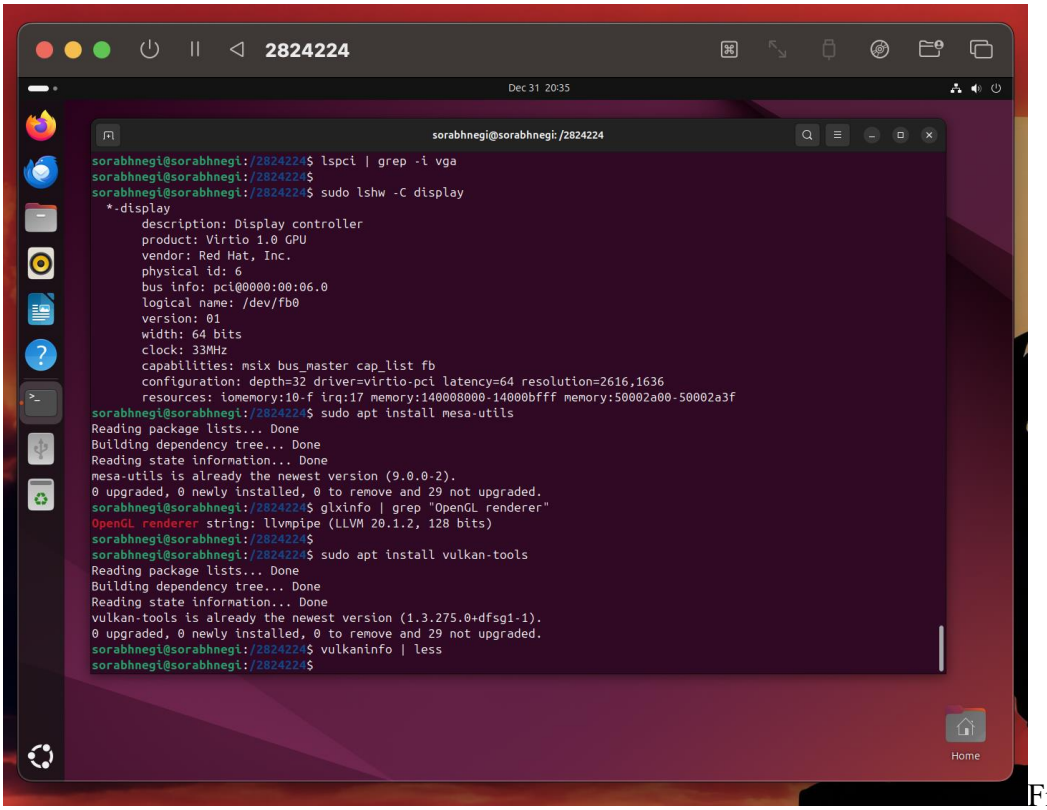


Figure 3

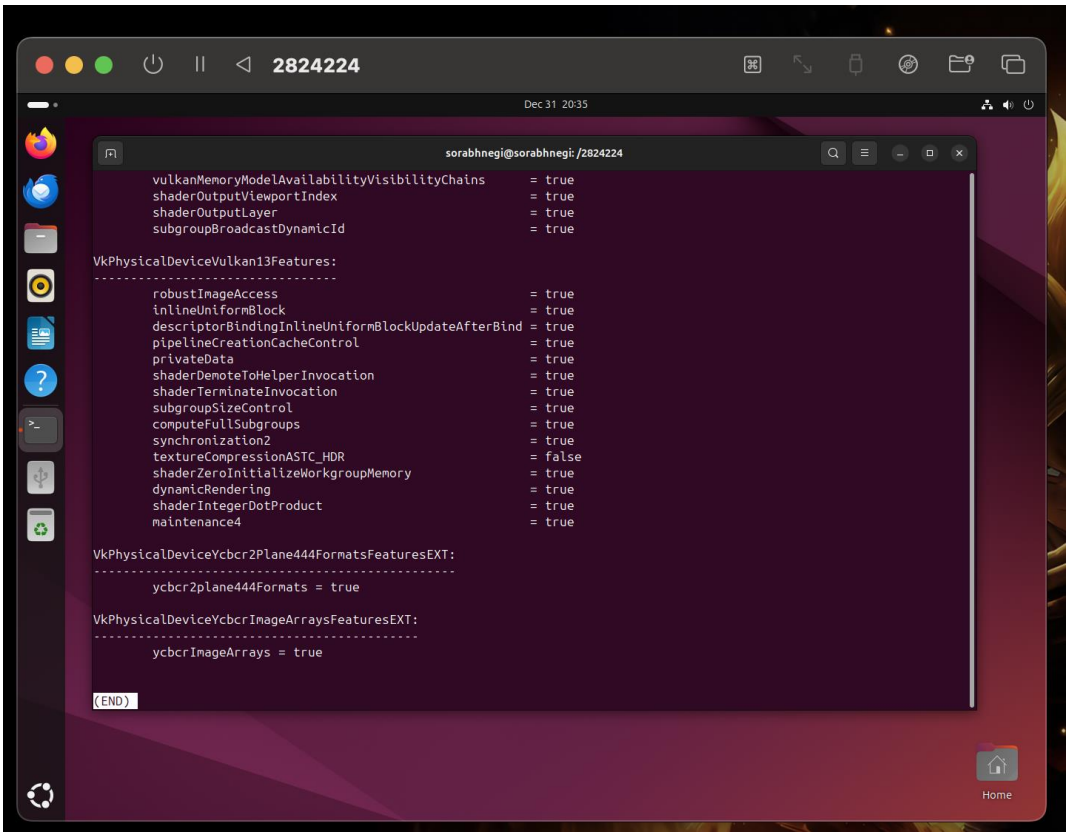


Figure4

4. **glxgears this is a simple GPU rendering demo to rectify if GPU acceleration works.**

This command shows how the GPU is performing in the system, and how it works in the system. Best for testing the performance of any GPU connected to the system.

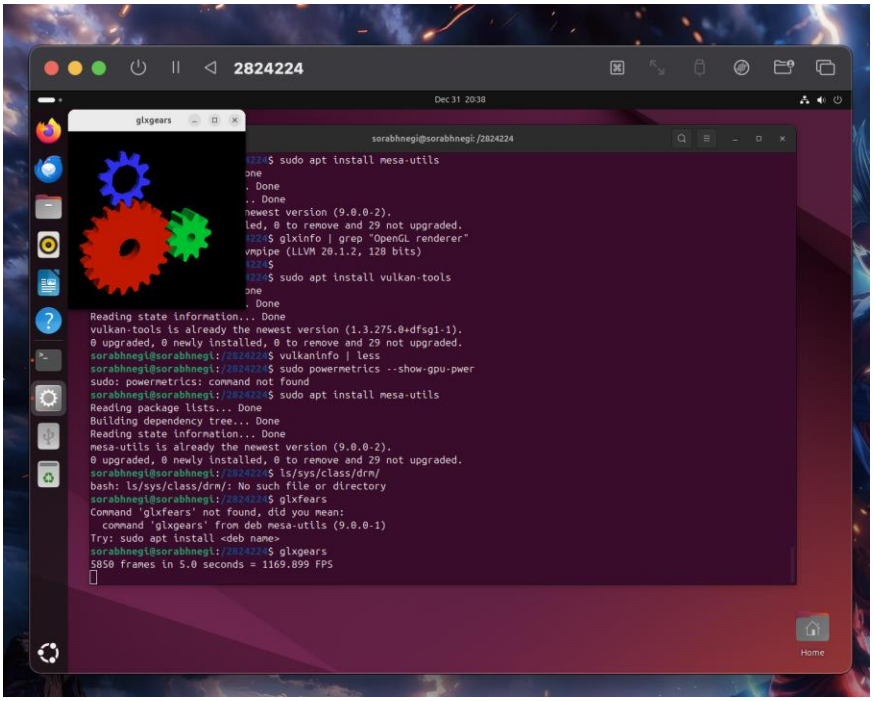


Figure 5

5. Vainfo If not installed run another command to install it sudo apt install vainfo.

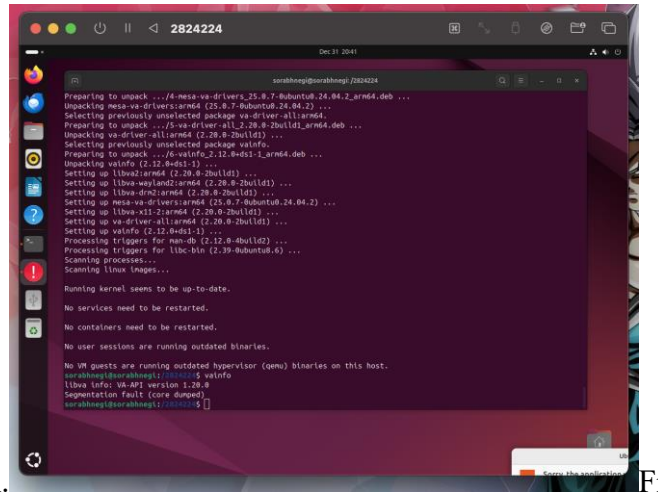
If the va info is not running so we must install it before executing the command with the use of another installation command to install it, then it will run smoothly to check Va information on the system. 

Figure 6

COMPARE AND CONTRAST AZURE AND AWS CLOUD PLATFORM

Comparison	AWS	Azure
Compute Services	Aws has an EC2 (elastic cloud compute) which offers various instance types optimized for specific purposes and GPU workloads. Also, some other serverless services like lambda, ECS and EKS.	Azure has Azure Virtual machines which are integrated to windows operating systems. Other services are Azure functions, Aks and VM scale sets.
Storage services	AWS has storage services, for example Amazon S3 (object storage). Some more are EBS, EFS, and Glacier.	Azure has storage services like Azure Blob storage, Disk storage, Azure Files and Azure Archive Storage.
Databases	Some services for Relational databases are RDS, aurora and for NoSQL is DynamoDB, and for DATA warehouse RED shift	In Azure for relational database we have Azure SQL database, for NoSQL there is Cosmos DB and last data warehouse there is Synapse Analytics.
Networking	Aws works on Virtual Private Cloud (VPC) for isolated networks. Load balancers are ELB and DNS handled by Route 53	Azure based on Virtual networks incorporation into premises in Microsoft environment. While load balancer is azure load balancer. DNS maintained by Azure DNS
Security and Identity Management	Aws IAM (identity and access management) with fine grained policy controls.	Azure has Azure Active Directory (Azure AD) which is responsible for identity access in cloud system
Pricing conditions	AWS uses pay as you go with reserved instances. More complex when it is about to predict cost. Discounts are reserved and spot instances.	Even Azure uses pay as you go, reserved VM instances and offers cost benefits for previous license holders.
Hybrid/Multiple cloud support	AWS has facilities for hybrid services like AWS outposts.	Azure has Arc which is consistent in management across on-prem and multi-cloud environment.

Developers' tools	**Aws offers various services like Pipeline, code build and cloud formation for the development functions**	**Azure has its own Azure DevOps and GitHub actions which are owned by Microsoft itself.**

Summary of the comparison

Aws and Azure both have performed well in past times, that's why they are doing great in the marketplace, but some points which make the difference are how users want to use them. So, most of the favors depend on the users how they wanted to use it. Meanwhile, comparison makes it easier to make choosing the required service for the implementation of cloud environment. I choose Azure for further demonstrating the functionality of cloud services. The deployment of resources groups is not easier in azure, but it provides more options and flexibility rather than amazon web services (AWS). According to my understanding, I find it comfortable to use Azure for my report.

Demonstration of the functionality that was implemented in Azure.

Creating a resource group and IAM in Azure

Starting with making an azure account provided by the university after creating a successful account. Sign in Azure with the credential portal.

https://azure.microsoft.com/en-us/free/students/ When you log in to the Azure cloud platform, you will see a lot of options but at the top of the page you will see a search box. Follow the command written below.

1. Type virtual machine in search box. You will see list of services

2. Choose Virtual Machines from Services. Then a page of the virtual machine will open.

3. Select Create from the Virtual Machine page. It will take you to a manual page to fill the requirements you need for the virtual machine.

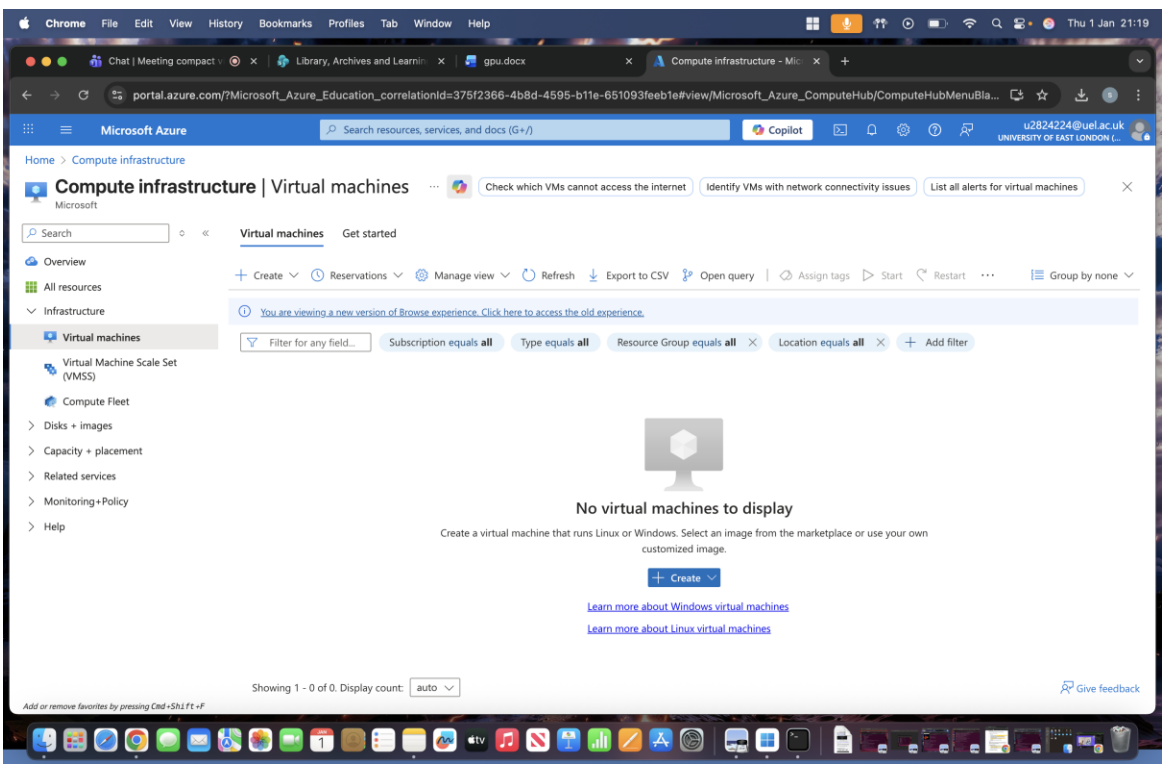


Figure 7

4. Choose Subscription as shown in Figure Azure for Students. That will not cost you for services because it's free and funded by azure.
   Resource group sorabhnegiu2824224
   And other instances details as shown in figure.
   Virtual machine name sorabhnegiu2824224
   Region Switzerland north
   Availability options availability zone

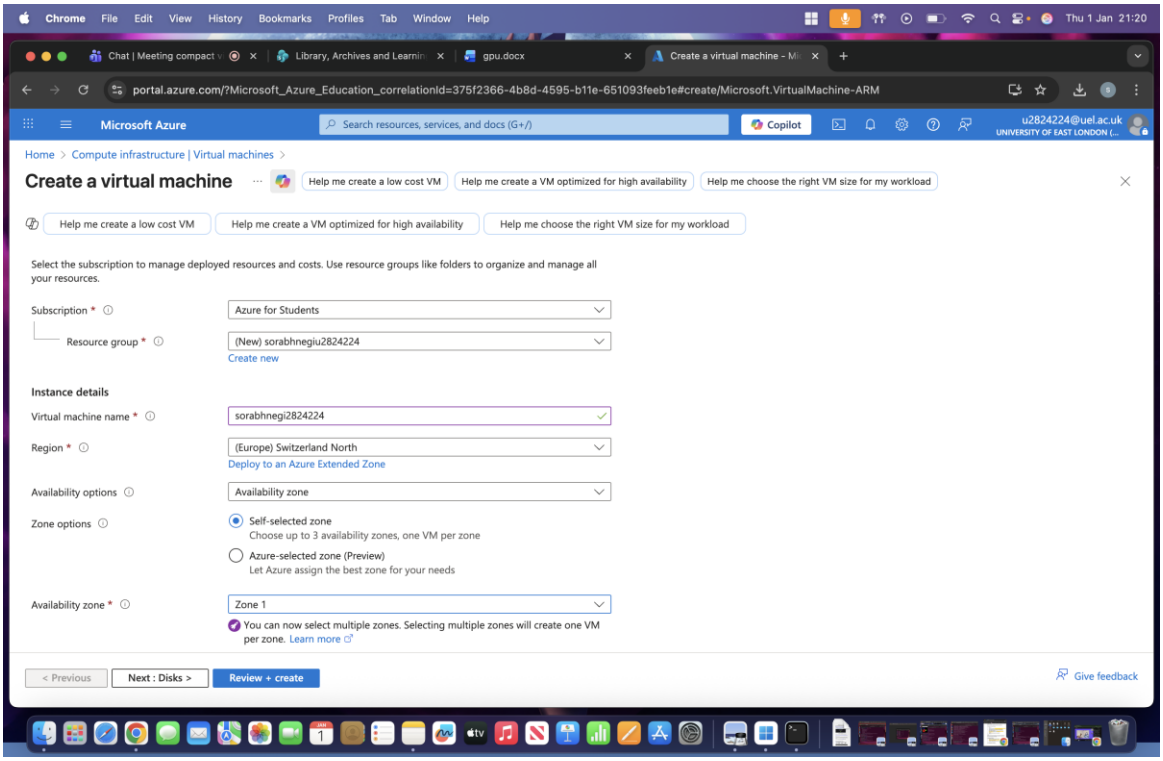


Figure 8

5. Account details to provide username sorabhnegiu2824224 and password Sora@123 to use it in the RDP file in upcoming steps to open the file. Remember the logins otherwise it will be a problem.

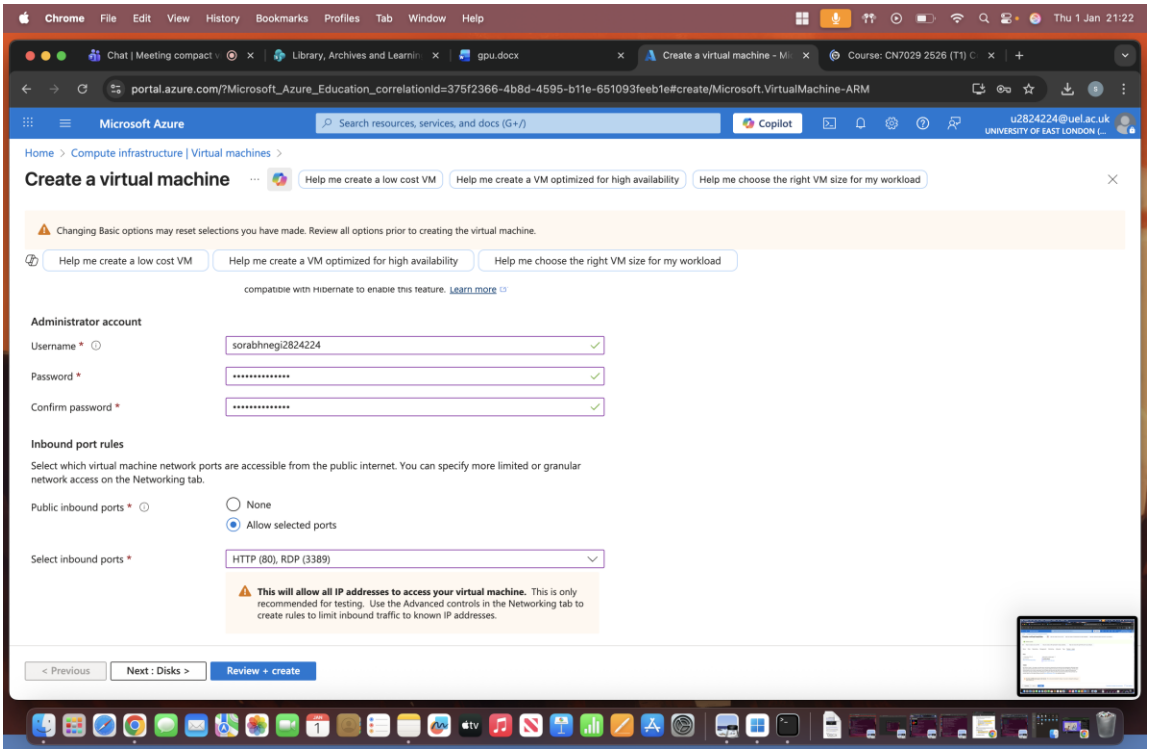


Figure 9

6. For Inbound port rules, Select HTTP 80 and RDP (3389) on allow selected ports. Https and RDP enable communication in cloud and network computing.
7. Select Review + Create for the creation. After this you will see a basic check for the deployment, and you can make changes here if needed.

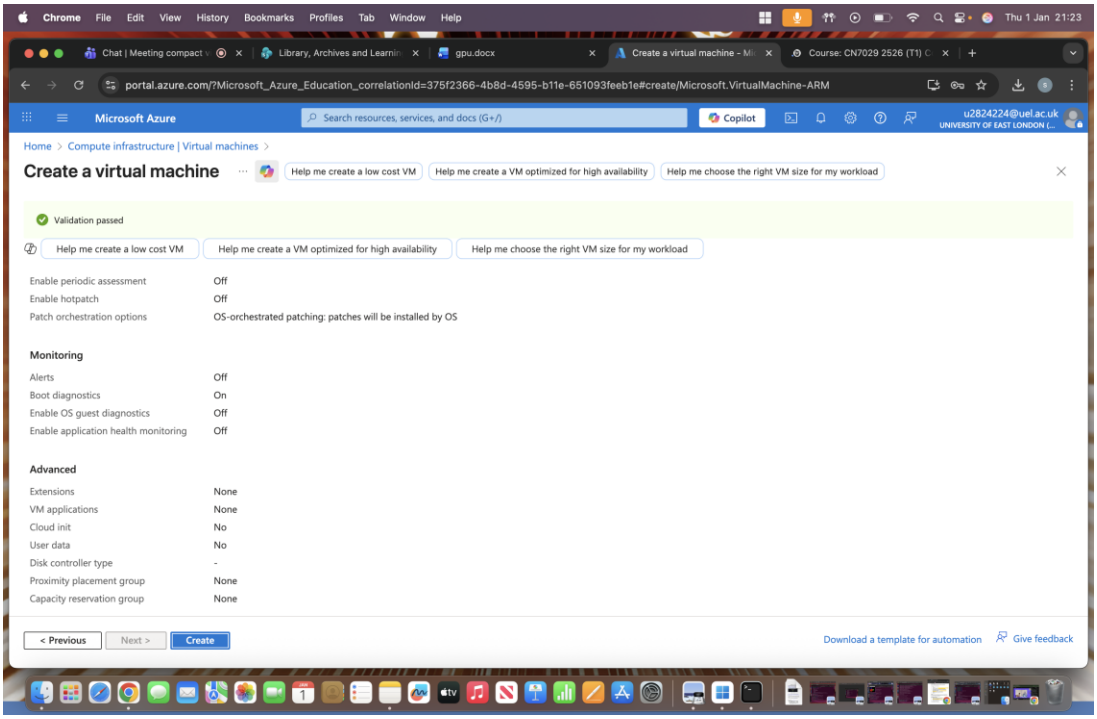


Figure 10

8. Select the Create button. Once you see the validation pass on the top of the screen, it's good to go for the creation of the virtual machine.
9. After deployment is done, Click on Go to resource. There is a resource created on the list.

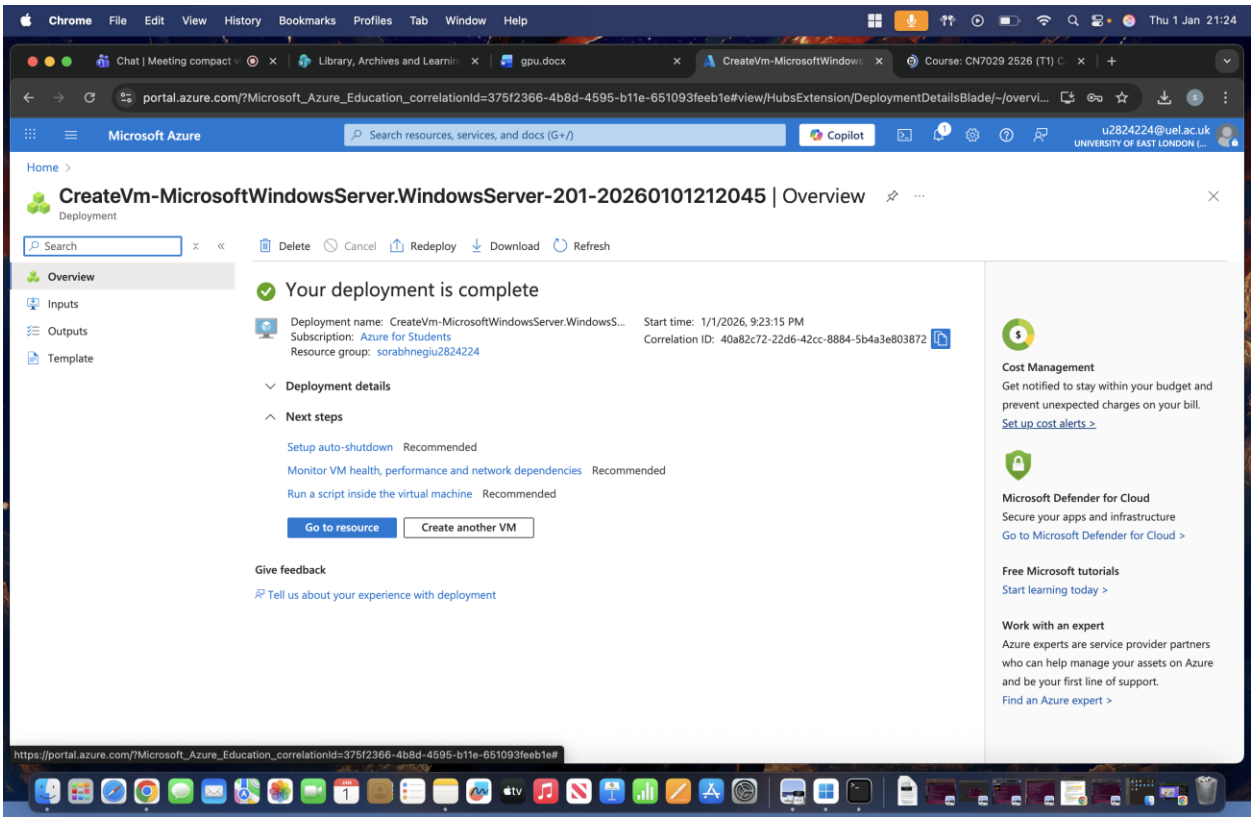


Figure 11

10. Now connect to virtual machine and download RDP file in your system to check the file.
11. In mac to open RDP file, we must use windows app in mac m3. Used the credentials which we used during the above steps.

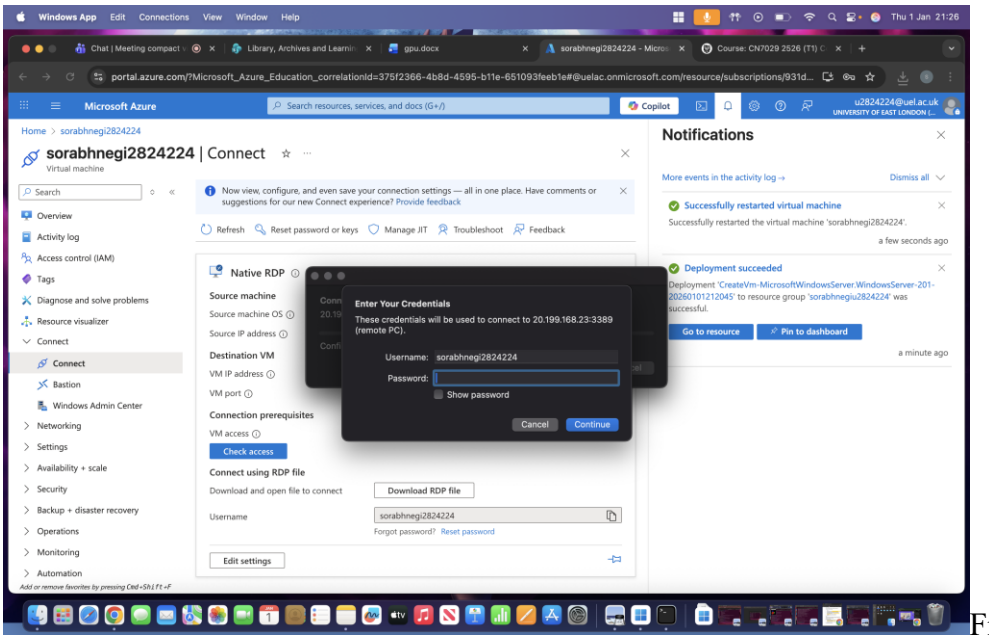
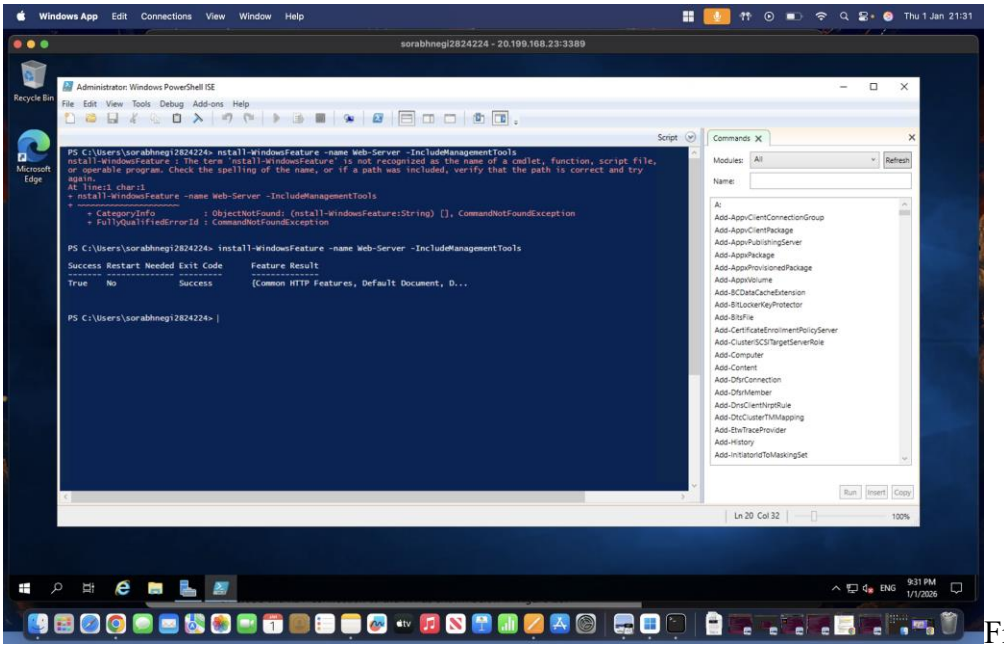


Figure 12

12. Now open the file and feed the password your virtual machine will runOpen power shell Install-WindowsFeature –name Web-Server –IncludeManagementTools

This command in powershell install the feature as shown in figure.


Windows App Edit Connections View Window Help
Thu 1 Jan 21:31
sorabhnegi2824224 - 20.199.168.23:3389
Recycle Bin
Microsoft Edge
Administrator: Windows PowerShell ISE
File Edit View Tools Debug Add-ons Help
PS C:\Users\sorabhnegi2824224> nstall-WindowsFeature -name Web-Server -IncludeManagementTools
nstall-WindowsFeature : The term 'nstall-WindowsFeature' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ nstall-WindowsFeature -name Web-Server -IncludeManagementTools
+ ~~~

This is the whole process of creating a resource group and IAM in Microsoft Azure. Steps are clearly illustrated above how to do it efficiently in the azure portal. The Azure portal provides a user interface that allows users to create, configure, and manage cloud resources effectively. One of the basic functionalities of deploying the services on azure is creating resource groups. The Identity and Access Management (IAM) in Azure is implemented through the Azure function. IAM enables personal secure access to role users' files.

EVALUATION

A Reflective evaluation of how Linux GPU Operating Systems command-line tools are used in Cloud Computing. After doing this coursework I got an understanding of the Linux command line tools related to GPU. This practical usefulness of Command-line tools is beneficial in monitoring, and optimizing GPU resources, specifically in cloud-based infrastructures with graphical user interfaces.

The Linux is the most used operating system in cloud computing. Azure intensively based on Linux-based virtual machines for workloads, for examples machine learning, analysis and simulator applications. For Linux is beneficial while using on cloud platforms because they majorly accessed remotely via SSH, RDP and HTTP. Also, for System admin method interaction. Meanwhile, the Linux operating system requires some skillsets of users to implement its benefits. The user with required skills suits best for the command line interface for Linux.

During the Linux command implementation, on terminal I run commands for GPU which provides utilization, memory usage, gear information, and power consumption. These are vital in cloud computing where GPU are essential and expensive. The facilities in Linux to check the ability to monitor GPU performance from the command line interface enables DevOps to ensure detect errors and reduce unnecessary errors. The most important part of every project is to manage its tasks, and the command line interface of Linux is most suitable for it.

I also recognized the generalization of GPU hardware in cloud environments. Cloud users do not have access to physical GPUs. Linux command-line tools can fulfill this by exposing hardware information through commands. This abstraction makes users optimize workloads. The information makes the cloud engineers sure for the availability of the hardware in the cloud, and they can take advantage of the functionality of the specification provided by the cloud application remotely.

However, all research and work have some limitations. The Linux command-line tools can be challenging for novices, somehow when GPU behavior differs across hardware architectures or virtual machines environments. This demonstrates the requirements for strong skills of Linux. But learning them with dedication and hard work is fruitful for longer times.

These Linux commands help me to improve my system problem solving ways to get the services faster than other methods. Troubleshooting GPU issues, compatibility, or performance rather than other feedback is much better in Linux. Knowing every aspect of the Gpu connected to the system enhances the performance of the system to the full extent and executes the task in seconds.

This experience demonstrated that the commands used in Linux for GPU are not the basic utilities but crucial enablers of cloud computing. Knowledge and learnings are helpful for the usefulness of clouds.

CONCLUSION

The coursework focused on learnings related to GPU oriented Linux commands line tools and a comparison between two cloud platforms which are Microsoft Azure and AWS. Throughout the work, it was explained that cloud computing platforms like Azure provided its own features like scalability, flexibility and pay as you go to services solutions for latest users' demands. Meanwhile, AWS is also a good example of cloud computing services, but both differed in terms of enterprises models, services maturity and ease of use. Therefore, there was a comparison and contrast between both services to show the core discrepancies between them.

Furthermore, the coursework has detailed in the Linux played a crucial role in cloud environments, especially for GPU to enable workloads. Because of the ownership of cloud GPUs, users can easily check and access their services using the command line interface. The tools are clearly efficient for resource utilization, supported automation, and stability in cloud platforms. These tools made command line interface more useful for users to make it possible to check their services in cloud platforms to manage their applications accordingly.

The making of the Azure resource groups and identity access management is the clear depiction of the cloud platform functionality in real time servers. These options are provided for users to make it beneficial rather than having private platforms. The main reason why major companies these days rely on the cloud computing platform.

In summary, this coursework concluded that Linux OS added with cloud platforms command line tools are substantially improving the efficiency and reliability of GPU based cloud environments. The cloud computing functionality perfectly demonstrated to show the usefulness of cloud. The steps were there of every command used in the Azure. This will help in providing a better base for the further understanding of new learnings and practical application in cloud-based experiments and highly intensive computer hardware operations. This is a wonderful experience for me while doing the whole coursework on cloud computing and research related to the command for GPU on Linux based operating system.

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